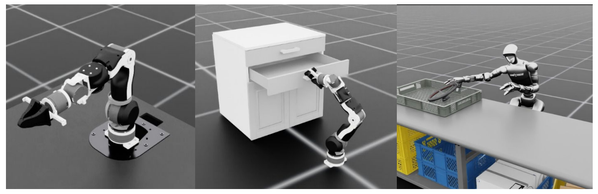
Dream2Policy: Learning Robot Manipulation Policies from Generated Human Videos

Christopher Luey, Wensi Ai, Karthik Dharmarajan

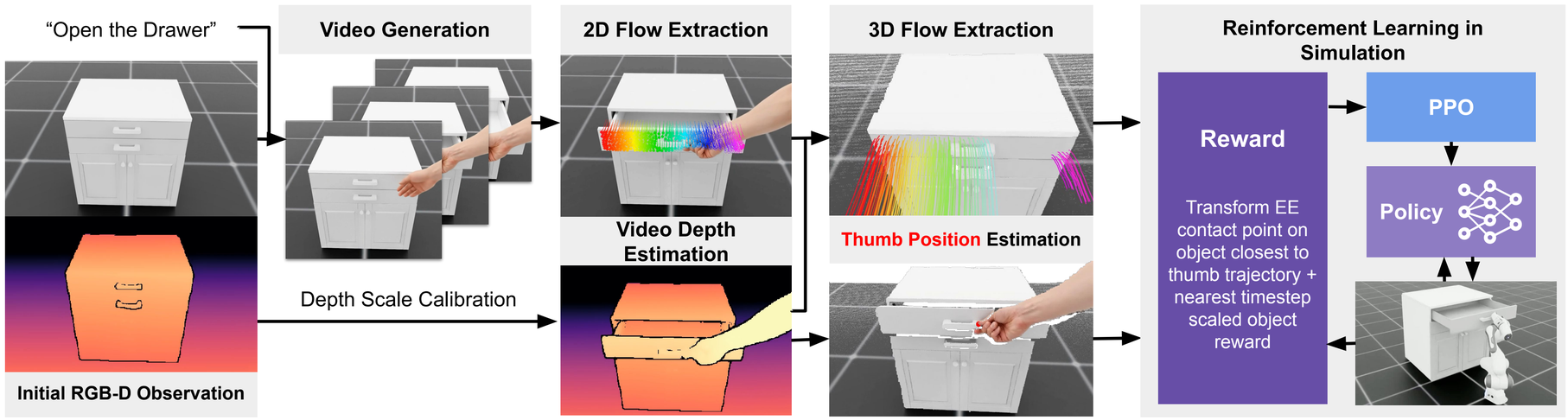
Problem



Learning general-purpose robot manipulation policies via RL is limited by the lack of accurate, scalable reward signals. Human demonstrations are expensive, and hand-crafted rewards require per-task engineering. We propose a framework that generates reward functions that are:

- Task agnostic
- Embodiment agnostic

Method



Background

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R)$$

Provide a suitable R from image of task and language instruction

- There are many approaches leveraging foundation models to provide reward functions, but we choose to **focus on video models**
- **Key Insight: Video generation models** can ground language instructions into **what needs to happen** and **how a human embodiment can achieve it**
- RL with the desired robot embodiment can benefit from this reward signal

	No expert data	No VLM in loop	Articulated objects
[1]VIPER	X	✓	✓
[2]Video2Policy	✓	X	✓
[3]Singh et al.	✓	✓	X
Dream2Policy	✓	✓	✓

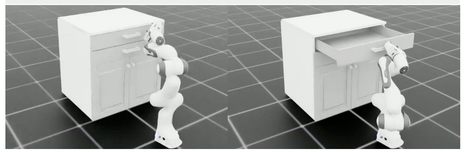
Experiments

Tasks

Lift Cube



Open Drawer

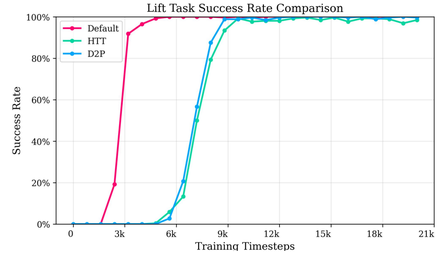


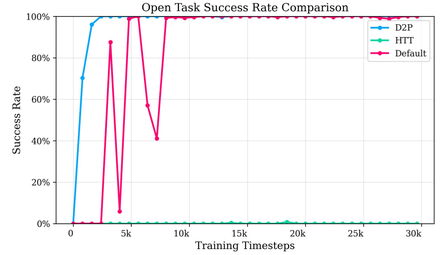
Baselines

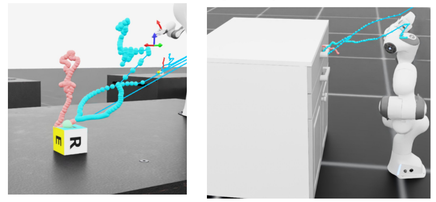
Default: Human hand engineered rewards

Hand Trajectory Tracking (HTT): Encourage end-effector to be within a certain distance of thumb trajectory + match object pose trajectory end point

Dream2Policy (D2P): Contact term derived from object points and closest thumb point + nearest timestep scaled object reward







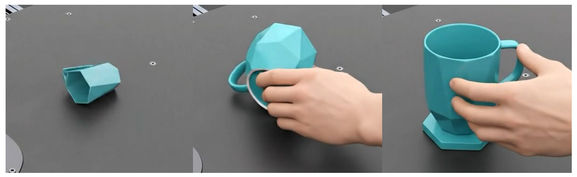
Visualization of thumb trajectory and object pose trajectory

Analysis

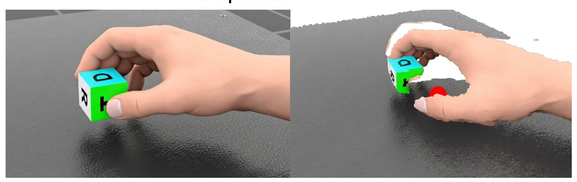
- While Dream2Policy ends up achieving close performance, it takes more training timesteps on Lift, due to the lack of task-specific reward terms

Failure Cases

Video Generation: Object Morphing



Depth Estimation



Original Frame — Incorrect Depth & Thumb Position

Conclusion

- Dream2Policy can successfully specify rewards for robot manipulation tasks from generated videos
- **Future work**: Real world extension, investigate dexterous hand domain, better hand and object flow consistency

[1] A. Escontrela, A. Adeniji, W. Yan, A. Jain, X. B. Peng, K. Goldberg, Y. Lee, D. Hafner, and P. Abbeel. Video prediction models as rewards for reinforcement learning. Neural Information Processing Systems, 2023.
[2] W. Ye, F. Liu, Z. Ding, Y. Gao, O. Rybkin, and P. Abbeel. Video2Policy: Scaling up manipulation tasks in simulation through internet videos, 2025. URL https://arxiv.org/abs/2502.09886.
[3] H. G. Singh, A. Loquercio, C. Sferrazza, J. Wu, H. Qi, P. Abbeel, and J. Malik. Hand-object interaction pretraining from videos. In International Conference on Robotics and Automation (ICRA). IEEE, 2025.